

Experiment-2

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Design a UI where users recall visual elements (e.g., icons or text chunks). Evaluate the effect of chunking on user memory.

FRAME 1:

INSTRUCTION PAGE:

Chunking Analysis of the Instruction Page

Chunking is a cognitive strategy that breaks information into smaller, manageable units, making it easier to process and remember.

The **Memory Recall Task – Traffic Theme** instruction page effectively applies chunking in the following ways:

1. Clear and Sequential Numbering:

- Instructions are presented in **six clearly numbered steps**, guiding users step-by-step.
- Each instruction focuses on **one action**, reducing confusion and cognitive load.

2. Logical Grouping of Information:

- **Observation Phase (Steps 1–2):** Explains what traffic signs or road icons will be shown.
- **Memorization Phase (Step 3):** Encourages users to focus on remembering as many signs as possible.
- **Recall Phase (Steps 4–5):** Describes how and when users will recall the items.
- **Performance Focus (Step 6):** Emphasizes accuracy over speed.

3. Visual Hierarchy and Design Elements:

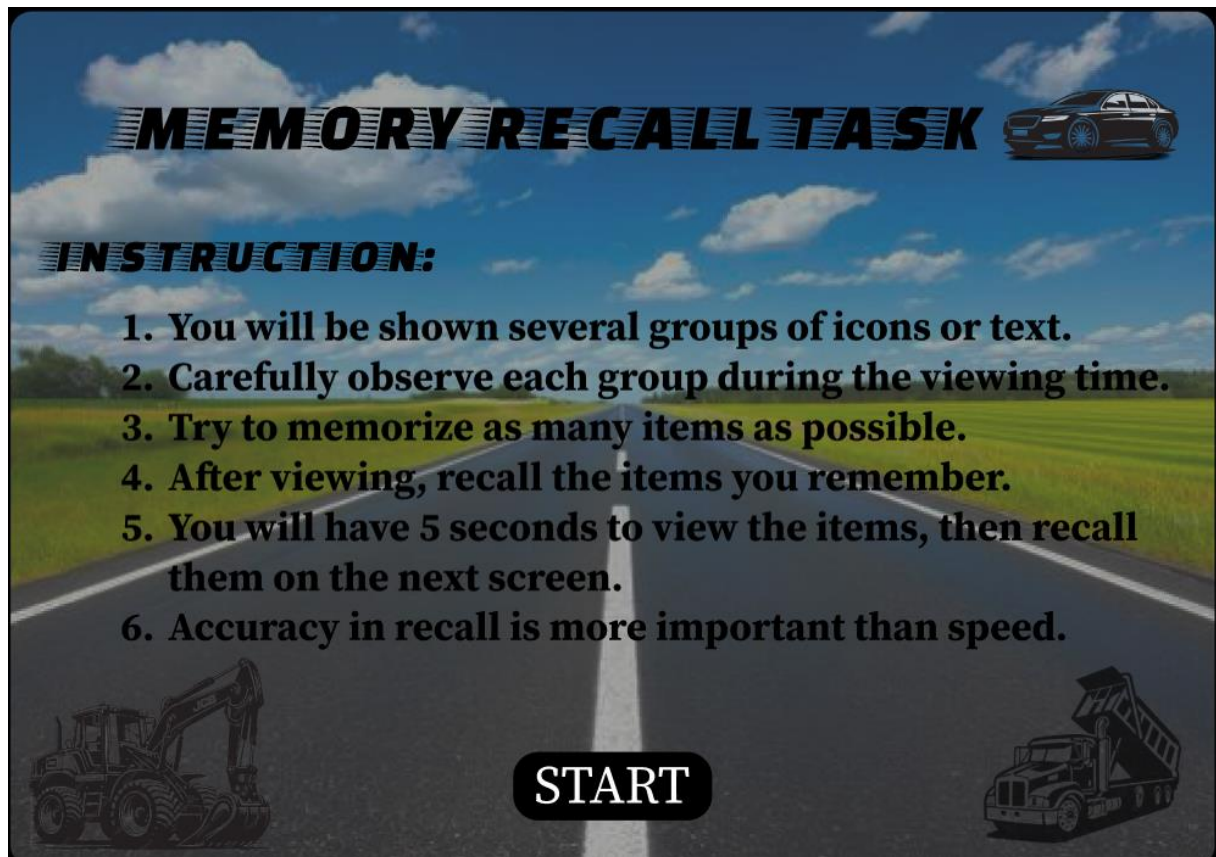
- **Bold heading “MEMORY RECALL TASK”** immediately communicates the task.
- **Road, highway, and vehicle-themed background** improves contextual relevance.
- **High-contrast START button** acts as a clear call-to-action.
- **Traffic and racing visuals** increase engagement without distracting from the task.

4. Time Constraint Reinforcement:

- The instruction **“You will have 5 seconds to view the items”** sets clear expectations.
- Users are mentally prepared to focus quickly.

5. Simplicity and clarity:

- Instructions use **short, direct sentences**.
- Active voice improves readability and task understanding.



FRAME 2:

CHUNKING PHASE:

Analysis of the Memory Recall Task - Chunking Phase Screen (Traffic Signs Theme)

This screen represents the visual memory encoding phase, where users observe and memorize different traffic signs and road symbols within a limited time.

1. Purpose of the Screen

- Allows users to **view and mentally group traffic signs** before recall.
- Encourages chunking by categorizing signs based on:
 - Warning signs
 - Mandatory signs
 - Prohibitory signs
 - Informational signs

2. Key UI Elements:

Countdown Timer (00:05)

- Indicates limited viewing time.
- Creates urgency and improves focus.

Progress Bar

- Visually represents remaining time.

- Reinforces time pressure intuitively.

Grid of Traffic Icons

- Displays multiple traffic signs in a structured grid.
- Encourages chunking strategies such as:
 - Grouping by shape (circle, triangle, square)
 - Grouping by color (red, blue, white)
 - Grouping by function (warning, restriction, information)

Vehicle/Road-Themed Graphics

- Enhances realism and relevance.
- Maintains engagement without cognitive distraction.

3. How Chunking Works:

- Users scan the icons and identify meaningful groups.
- Time limitation forces quick pattern recognition.
- Users prepare for the recall phase.

4. How the Chunking Phase Works:

- Improves short-term memory retention.
- Reduces mental overload.
- Enhances visual categorization skills.



FRAME 3:

RECALL PHASE:

Analysis of the Memory Recall Task - Selection Phase

This screen represents the memory retrieval stage, where users recall and select the traffic signs they previously viewed.

1. Purpose of the Screen:

- Tests how effectively users retained visual information.
- Measures the success of chunking strategies.

2. Key Elements and UI Components:

Title & Instructions

- "MEMORY RECALL TASK"
- "SELECT THE TRAFFIC SIGNS YOU REMEMBER"

Selection Grid

- Includes both:
 - Signs shown in the chunking phase
 - New distractor signs
- Users select remembered signs using selection indicators.

○ Selection Indicators

- Help users clearly confirm their choices.

🚗 Road-Themed Background

- Maintains consistency across screens.

3. How the Selection Phase Works:

- Users compare displayed signs with memory.
- Select remembered signs.
- Distractors test memory accuracy.
- SUBMIT button finalizes choices.

4. Cognitive and UX Benefits:

- Evaluates true recall vs false memory.
- Clear interaction improves usability.
- Gamified traffic theme maintains motivation.



FRAME 4:

RESULT PAGE:

Analysis of the Memory Recall Task - Score & Feedback Screen

This screen presents the performance feedback based on user recall accuracy.

1. Purpose of the Screen:

- Displays recall score clearly.
- Allows users to choose next action.

2. Key Elements & UI Components:

Score Display

- **"YOUR RECALL SCORE"**
- Digital scoreboard (e.g., 5/5)
- Racing flag graphics symbolize completion.

Action Buttons

- **CONTINUE – Move to next level**
- **RESTART – Retry the task**
- **EXIT – End the task**

Vehicle-Themed Visuals

- Reinforce speed, focus, and performance metaphors.

3.How This Phase Works:

- System evaluates selections.
- Calculates recall accuracy.
- Displays results visually.
- User chooses next action.

4.Cognitive & UX Benefits:

- Immediate feedback improves learning.
- User control increases engagement.

- Competitive racing theme makes evaluation enjoyable.



PROTOTYPE LINK:

<https://www.figma.com/proto/uBWZZBzHXciHfsQB1GMXsC/Untitled?node-id=13-2&t=uBtu1JdkgXINksRc-1>

